The economics of architecture

Gabriel M. Ahlfeldt
Humboldt University
Elisabetta Pietrostefani
University of Liverpool
Ailin Zhang
LSE

05/2026

Context

- **Distinctive architectural design** Examples
  - **Aesthetic features** that provide utility beyond that the mere use of the building
  - Similar to **consumption value of arts**
- **Internal value** to residents of a distinctive building
  - Reflected in rents to **owner of distinctive building**
- **External value** to neighbors (in addition to society as a whole)
  - Mostly reflected in rents to **owners of other buildings**
- **Coordination problem**
  - Private cost vs shared benefit incentivizes **freeriding**
- Scope for **welfare-improving planning policies**
  - Subsidies to owners implementing distinctive design
  - Zoning (distinctive districts, FAR incentives)

This paper

- **Develop a quantitative model** for the analysis of design policies
  - Downward-sloping demand for distinctive design meets constant marginal cost
  - Design spillover shifts location-specific amenity
- **Review evidence** to guide quantification of the model
  - Internal design premium: 15%; External design premium: 9%
- **Survey** to understand design preferences
  - Buildings rated highly by Starchitect are valued higher by respondents
  - Dispersion of tastes for architecture similar to location preferences: $\epsilon = 4$
- **Simulate design policies**
  - Optimal Subsidy (10% of construction cost) improves welfare
  - Design districts can generate small improvements when they are small
  - FAR incentives require strong constraints that depreciate welfare
  - Super developers trade internalization of spillovers against monopoly rent

Related literature

- **Design premiums**
  - e.g. Hough and Kratz (1983); Vandell and Lane (1989); Ahlfeldt and Holman (2018); Koster et al. (2016); Lindenthal and Johnson (2021); Füß et al. (2025)
    - Our quantitative survey covers 68 estimates of premiums from **41 studies**
    - Our quantitative survey covers 33 estimates of external premiums from **28 studies**
- **Quantitative urban models**
  - e.g. Ahlfeldt et al. (2015); Heblich et al. (2020); Tsivanidis (2025); Redding (2025)
    - We focus on choices among buildings of different design within a neighborhood
- **Welfare effects of planning**
  - e.g. Glaeser et al. (2005); Brueckner and Rossi-Hansberg (2010); Sridhar (2012); Turner, Haughwout, van der Klaauw (2014); Cheshire and Dericks (2020); Ahlfeldt and Barr (2022)
    - Design choice by the developer is endogenous in our model

Model

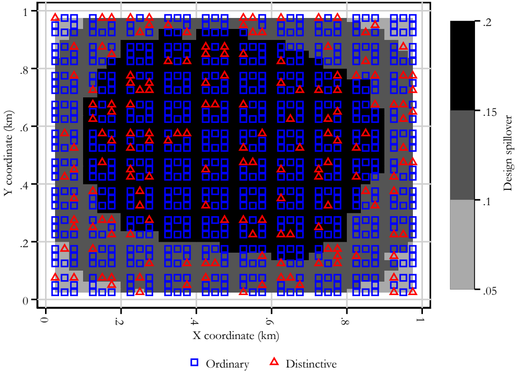
- **Geography:** Neighbourhood with J parcels and one developer per parcel
  - Each parcel has developable land $\bar{K}_i$, can be develoed by one building
  - Embedded in wider city
- **Residents:** Choose **a building** (either distinctive or not) or the rest of the city
  - Standard Cobb-Douglas preferences for housing and floor space
  - Amenity B_i includes exogenous component and **design spillovers** D_i
    - Take the standard market potential form (normalized to range between 0 and 1)
  - Idiosyncratic preferences (Fréchet) for submarkets
    - **Downward-sloping demand** within submarkets: Distinctive, ordinary, rest of city
- **Developers:** Choose building height h_i and **design** d
  - Face design-specific construction cost and height elasticity of cost θ
    - Heterogeneity in design cost generates **upward-sloping aggregate supply**
  - May receive subsidy t_i^d and face design regulation G_i

Primitives and endogenous outcomes

- **Equilibrium:**
  - Utility equalizes within neighbourhood
  - Zero profits in floor space production
  - Floor space markets clears
- **Primitives:**
  - Calibrated/estimated parameters, neighbourhood wage, utility in rest of city
  - Parcel-specific fundamental amenity, design subsidy, height limit, design restriction
  - City population, parcel land
- **Endogenous outcomes:**
  - Submarket population, expected city utility, developer profits
  - Parcel rent, height, floor space consumption, distinctive design status & spillover
- **Mapping from primitives to endogenous outcomes**, solved numerically

Workers Developers Welfare Equilibrium

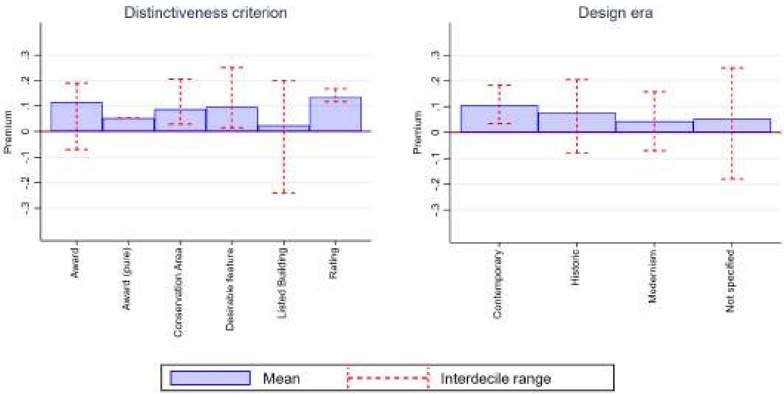
Intuition and quantification



- Developers develop distinctive buildings **where it is profit-maximizing**
- **Marginal parcel** is where the return to distinctive design falls below the cost
- Exposure to distinctive design **spillover translates into higher prices**
- **Need moments to calibrate the model**
  - Internal design premium
  - External design premium
  - Taste dispersion

Internal design premium

- Literature search reveals 68 estimates from 41 studies



Internal design premium

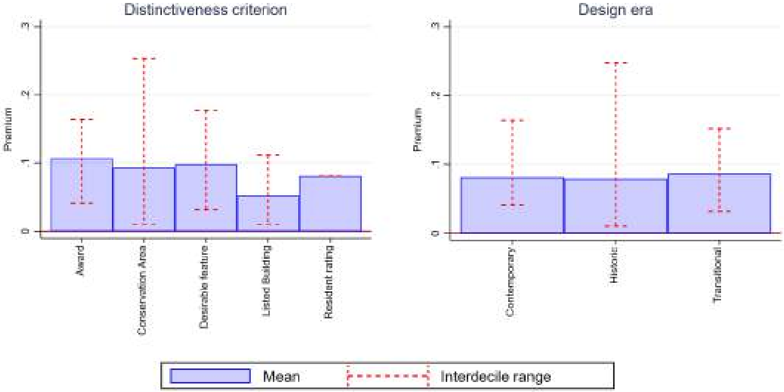
	(1) Premium		(2) Premium		(3) Premium	
Commercial	0.095***	(0.02)	0.101***	(0.02)	0.146***	(0.02)
Residential	0.077***	(0.02)	0.099**	(0.04)	0.164***	(0.04)
Conservation Area			-0.02	(0.03)		
Listed Building			-0.08*	(0.05)		
Award (pure)			-0.05	(0.04)		
Historic Style			0.010	(0.05)		
Weighted	No		No		Yes	
Sample	All		All		Cohort control	
r2	0.4		0.4		0.7	
N	68		68		17	

Notes: Standard errors in parentheses. Each observation is an estimate of the effect of distinctive design on property price or rent from the literature in log points. All explanatory variables are dummy variables. Baseline distinctiveness measure is listed building. Modern covers the style groups contemporary, modernism, and transitional from Figure 12 (right panel). Quality weights are proportionate to Google citation counts, regression adjusted for publication years. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

- Regression-based average
  - We weight observation by publication-year adjusted citation count
  - Most credible estimates are from studies that identify the distinctiveness effect within cohorts
  - **Interal design premium $\approx$ 15 %**

External design premium

- Literature search reveals 33 estimates from 28 studies: **Average** ≈ **9%**



Design preferences

- **Internet survey** asking respondents to rate 10 buildings on a 0-100 scale
  - Also obtained ratings by "Starchitect" Stefano Boeri (Bosco Verticale, Milan)
- Structural estimation of **Fréchet shift and share parameters** using GMM: $\epsilon = 4$
  - Buildings rated highly by Starchitect also appreciated by respondents ($\rho = 0.8$)

Group	Mean rating	Variance	Shift A_b	Shape ε
All respondents, all buildings	52.72	480.88	42.73	4.04
People similar to Starchitect	51.34	416.81	39.53	4.06
People different from Starchitect	52.99	586.57	42.16	3.79
Buildings ranked high by Starchitect	65.63	486.86	55.89	4.69
Buildings ranked low by Starchitect	39.80	474.90	32.39	3.40

Notes: This table reports GMM estimates based on matching the first and second moments of the building-level rating distributions to those implied by a Fréchet model. The shape parameter ε governs preference heterogeneity, while A_b reflects the building-specific location shift. Groups are defined based on respondent-level or Starchitect-level ranking similarity.

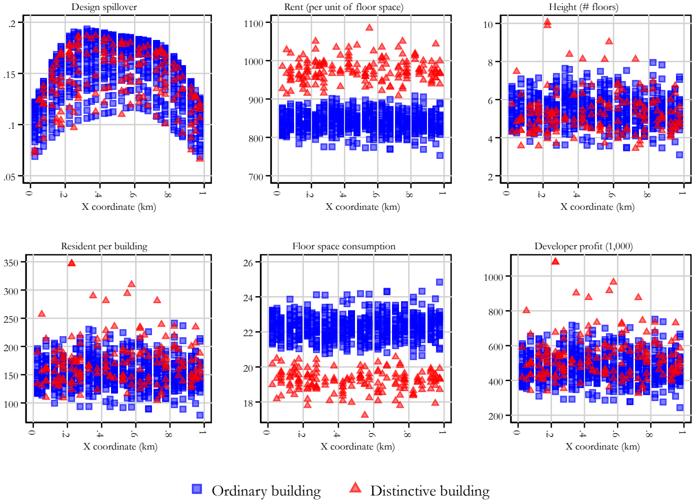
Examples GMM

Quantification

- **Borrow canonical parameters** from literature
  - Housing expenditure ($1 - \alpha = 0.33$), spillover decay ($\tau = 5$), height elasticity of construction cost ($\theta = 0.2$)
- **Estimate taste dispersion** using survey data
  - $\epsilon = 4$
- **Structural interpretation** of external design premium (for given $1 - \alpha$)
  - Normalize design spillover so that 1 reflects a fully distinctive neighbourhood
- Use **SMM** to estimate the average taste for and cost of distinctive design
  - The internal design premium via Frechet scale parameter
  - The average cost of distinctive design via target share of distinctive buildings (20%)
- This approach **works for real-world neighbourhoods**, too!

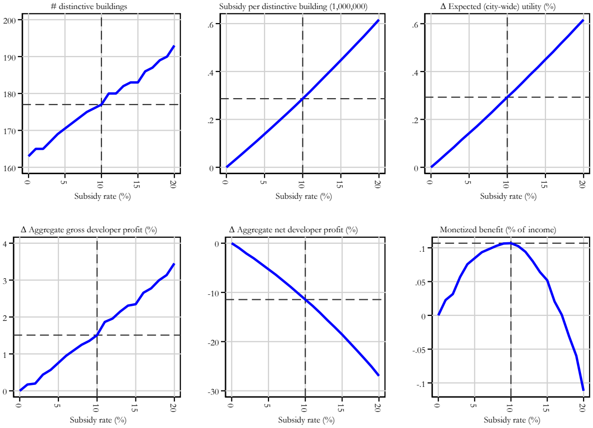
SMM

Neighbourhood structure



- Distinctive design premium
  - Matched to data
- Affects floorspace consumption
  - Higher price ⇒ lower consumption
- Design spillover is greatest in the centre
  - Affects prices (effect barely visible)

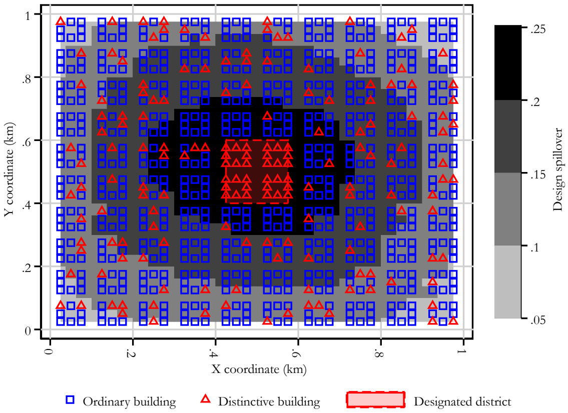
Pigovian subsidy



- Increases
  - # distinctive buildings
  - Expected utility
  - Developer gross profits
- Reduces net profits after taxes
  - Convex shape
- Hump-shaped welfare effect
  - Monetized utility effect, developer profits, and tax cost
  - Downward-sloping demand $\Rightarrow$ concavity
- **Optional subsidy**:
  - 10% of construction cost
  - $\approx \frac{1}{3}$ of distinctive design cost

Subsidy gradients

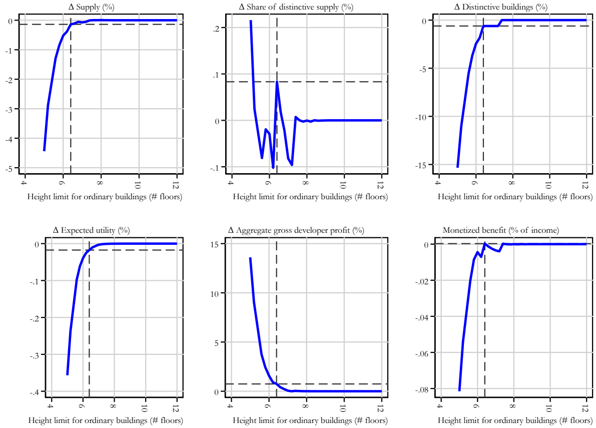
Mandatory distinctive design district



- Can **force developers** to solve the coordination problem
  - All buildings in a district must be distinctive
  - Related policy: Conservation areas
- Can generate **benefits when the distinctive district is small**
  - Distinctive buildings +4.3%
  - Expected utility +0.003% (city-wide)
  - Developer profits +0.018%
- If it is too large, the effect is negative due to diminishing returns

Map large district | Gradient | Gradient: Large district

FAR incentives



- **FAR incentives** only work under **constraints** for ordinary buildings
  - Introduce height limit for ordinary buildings
  - No limit for distinctive buildings
- Generally, binding **height limits depreciate welfare**
  - Non-binding height limits do not affect distinctive supply
- **Knife-edge welfare gain**
  - At 6.4-foor height limit
    - hard to hit in practice
  - Tighter constraints depreciate aggregate welfare and transfare welfare from workers to landowners

Gradients

Super developer

- **Coordination via super-developer:**
  - Have an incentive to internalize spillovers
  - Also have an incentive to maximize monopoly rent from scarce, distinctive design...
- **Small positive effect on distinctive supply** compared to myopic developer
  - A strategic developer ignoring spillovers would supply much less distinctive design
  - Monopoly quantity setting partially offsets the incentive to internalize spillovers

Scenario comparison	Dist. build.	Dist. supply	Supply	Rents	Expected utility	Pop	Profits
Myopic → Strategic	+3.1%	+0.2%	-0.31%	+0.44%	+0.01%	+0.06%	+0.06%
Strategic → Strat. w/o spillover	-13.7%	-3.3%	+0.34%	-0.43%	-0.02%	-0.13%	-0.12%

Notes: The table reports equilibrium outcomes under three super-developer scenarios using the greedy algorithm. Under the *fully-myopic* scenario the super developer make sequential parcel choice based on fixed rents and historical spillover. In the *fully-strategic* developer is monopoly who anticipate current equilibrium spillover. In the *strategic without spillover* scenario, developer is monopoly with historical spillover.

Conclusion

- Distinctive architectural design generates substantial **positive externalities**
  - Under-internalized in laissez-faire markets
  - Market fails to deliver optimal design levels
- **Net social benefit** of additional distinctive design justifies **policy intervention**
  - **Most efficient:** Construction cost subsidy targeted at distinctive buildings
  - **Design standards** in moderate districts can leverage spillovers
  - **FAR bonuses** offer marginal gains and risk inefficiencies if miscalibrated
    - **Should not introduce supply constraints** to enable policy
  - **Super developers** trade spillovers vs monopoly rent
- Scope for future research
  - Interaction of distinctive design and urban context
  - Cost of distinctive design

Appendix

Ordinary

London | Barcelona | Bath | Milan

Distinctive

Context

Workers: Preferences and indirect utility

- Workers choose among submarkets:
  - $\mathbf{C} = \{\text{distinctive}, \text{ordinary}, \text{outside}\}$
- Utility:
  - $U(v) = \tilde{U}^d a^d(v)$
  - $a^d(v)$ captures idiosyncratic taste for submarket d
- Within-neighbourhood utility:
  - $\tilde{U}^d = B_i \left(\frac{g}{\alpha}\right)^\alpha \left(\frac{f_i^d}{1-\alpha}\right)^{1-\alpha}$
- Housing demand:
  - $f_i^d = (1-\alpha)\frac{\bar{w}}{Q_i^d}$
- Indirect utility:
  - $\tilde{V}^d = A^d B_i^d \frac{\bar{w}}{(Q_i^d)^{1-\alpha}}$

Workers: Choice probabilities and outside option

- Idiosyncratic tastes:
  - $a^d(v) \sim \text{Fréchet}(A^d, \varepsilon)$
  - Shape parameter ε governs taste dispersion
- Choice probabilities:
  - $\mu^d = \frac{(\tilde{V}^d)^\varepsilon}{\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon}$
- Outside option:
  - $\tilde{V}^{outside} = \bar{V}$
- Interpretation:
  - Higher amenity and lower rents increase market share
  - Larger ε implies more homogeneous preferences
- Key implication:
  - Generates downward-sloping demand for distinctive buildings

Workers: Design spillovers and local amenity

- Local amenity:
  - $B_i^d = b_i^d \cdot \exp[\beta D_i]$
- Design spillover:
  - $D_i = \frac{\sum_{z \neq i} \omega_{iz} \mathbf{1}(\tilde{d}_z = \text{distinctive})}{\max_i \left(\sum_{z \neq i} \omega_{iz} \right)}$
- Spatial weight:
  - $\omega_{iz} = (\bar{K}_z h_z) \exp(-\tau \mathcal{D}_{iz})$
- Interpretation:
  - Nearby distinctive buildings raise local amenities and rents
  - Larger nearby buildings generate stronger spillovers
  - Spillovers decay with distance
- Coordination problem:
  - Developers ignore benefits to neighbouring parcels

Back to Model

Developers: Technology and profits

- One developer per parcel i
  - Chooses building design $d \in \{\text{distinctive}, \text{ordinary}\}$
  - Chooses height h_i^d on parcel land $\bar{K}_i$
- Revenue:
  - $Q_i^d h_i^d \bar{K}_i$
  - Q_i^d is the unit price of floor space
- Construction cost:
  - $\bar{C}(h_i^d)^{1+\theta} \exp(\delta_i^d - t_i^d)\bar{K}_i$
  - θ governs how construction costs rise with height
  - δ_i^d is a design-specific cost shifter
  - t_i^d is a construction-cost subsidy
- Profit:
  - $\pi_i^d = \left[Q_i^d h_i^d - \bar{C}(h_i^d)^{1+\theta} e^{\delta_i^d - t_i^d} \right] \bar{K}_i$

Developers: Height and design choice

- Profit-maximising height:
  - $h_i^{d^*} = \left(\frac{Q_i^d}{(1+\theta)\bar{C}e^{\delta_i^d - t_i^d}} \right)^{1/\theta}$
- Height regulation:
  - $\tilde{h}_i^d = \min(h_i^{d^*}, \bar{h}_i^d)$
  - $\bar{h}_i^d$ is the design-specific height limit
- Design choice:
  - If $G_i =$ distinctive: planner requires distinctive design
  - Else: developer chooses
    - $\tilde{d}_i = \arg\max_{d \in \mathbf{D}} \pi_i^d$
- Floor-space supply:
  - $H_i^d = \tilde{h}_i^d \mathbf{1}(\tilde{d}_i = d)\bar{K}_i$
- Key mechanism:
  - Developers internalize own rents, but not spillovers on neighbouring parcels

Back to Model

Welfare

- Two welfare components:
  - Worker utility: $\mathbb{E}[U]\bar{N}$
  - Developer profits:
    - $\Pi = \sum_{i,d} \pi_i^d 1(\tilde{d}_i = d) + \bar{\pi}^{outside}$
- Expected utility:
  - $\mathbb{E}[U] = \Gamma\left(\frac{\varepsilon - 1}{\varepsilon}\right)\left(\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon\right)^{1/\varepsilon}$
- Welfare metric:
  - Utility changes converted into income-equivalent units
  - Cobb-Douglas structure implies utility scales proportionally with income
- Policy evaluation:
  - $\Delta W = \frac{\mathbb{E}[U]^1 - \mathbb{E}[U]^0}{\mathbb{E}[U]^0} \bar{w}\bar{N} + \Delta\Pi$
- Subsidy cost:
  - $S = \sum_{i,d} \mathbf{1}(\tilde{d}_i = d)\bar{C}(h_i^d)^{1+\theta}\bar{K}_i\left[e^{\delta_i^d} - e^{\delta_i^d - t_i^d}\right]$

Back to Model

Equilibrium: Spatial equilibrium conditions

- Workers are mobile:
  - Across neighbourhoods via outside option (constant utility)
  - Across submarkets within neighbourhood (constant expected utility)
- Developers are competitive:
  - Take rents and spillovers as given
  - Choose profit-maximising design and height
- Floor-space markets clear at parcel level:
  - $f_i^d N_i^d = H_i^d$
- Submarket populations are consistent with discrete choice:
  - $\mu^d = \frac{1}{\bar{N}} \sum_i N_i^d$
  - $\mu^d = \frac{(\tilde{V}^d)^\varepsilon}{\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon}$
- Spatial equilibrium:
  - Rents adjust such that demand equals supply across submarkets

Equilibrium: Recursive structure

- Conditional on:
  - $\{\tilde{d}_i, Q_i^d, \tilde{V}^{distinctive}, \tilde{V}^{ordinary}\}$
- Workers:
  - Housing demand: $f_i^d = (1-\alpha)\frac{\bar{w}}{Q_i^d}$
  - Choice probabilities: $\mu^d = \frac{(\tilde{V}^d)^\varepsilon}{\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon}$
- Developers:
  - Optimal height: $h_i^{d*} = \left(\frac{Q_i^d}{(1+\theta)\bar{C}e^{\delta_i^d - t_i^d}} \right)^{1/\theta}$
  - Feasible height: $\tilde{h}_i^d = \min(h_i^{d*}, \bar{h}_i^d)$
- Implied equilibrium objects:
  - Spillovers D_i, amenities B_i^d, supply H_i^d
  - Population N_i^d, utility $\mathbb{E}[U]$, profits Π

Equilibrium: Fixed-point problem

- Unknown equilibrium objects:
  - Design choices $\tilde{d}_i$
  - Rents Q_i^d
  - Submarket utilities $\{\tilde{V}^{distinctive}, \tilde{V}^{ordinary}\}$
- Submarket utilities pinned down by:
  - Population consistency: $\mu^d = \frac{1}{\bar{N}} \sum_i N_i^d = \frac{(\tilde{V}^d)^\varepsilon}{\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon}$
- Design choices pinned down by:
  - Profit maximisation: $\tilde{d}_i = \arg\max_{d \in \mathbf{D}} \pi_i^d$
- Rents pinned down by:
  - Parcel-level market clearing: $f_i^d N_i^d = H_i^d$
- Numerical solution:
  - Fixed point because spillovers affect both rents and design choices
  - Avoid multiplicity via convexification (heterogenous design cost)

Back to Model

Survey Examples I

40 (23)

48 (22)

29 (19)

39 (23)

61 (25)

75 (21)

72 (19)

76 (19)

Back to Preferences More

50 (24)

42 (21)

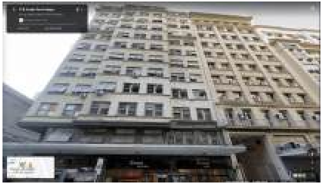
37 (21)

55 (25)

73 (19)

58 (23)

Back to Preferences back More



34 (21)

46 (24)

35 (20)

62 (22)

66 (24)

57 (23)

Back to Preferences back

Survey: Fréchet taste heterogeneity

- Survey data:
  - Respondents rate buildings on a 0–100 scale
  - Ratings interpreted as latent taste draws a_{ib}
- Structural assumption:
  - a_{ib} follows a Fréchet distribution
  - $F(a) = \exp\left(-(a/A_b)^{-\varepsilon}\right)$
- Parameters:
  - A_b: building-specific attractiveness
  - ε: taste heterogeneity across respondents
- Theoretical moments:
  - $\mathbb{E}[a_b] = A_b \Gamma\left(1 - \frac{1}{\varepsilon}\right)$
  - $\mathrm{Var}(a_b) = A_b^2 \left[\Gamma\left(1 - \frac{2}{\varepsilon}\right) - \Gamma\left(1 - \frac{1}{\varepsilon}\right)^2\right]$

GMM Estimation

- Empirical moments:
  - $\hat{\mu}_b = \frac{1}{n_b} \sum_{i=1}^{n_b} a_{ib}$
  - $\hat{\sigma}_b^2 = \frac{1}{n_b - 1} \sum_{i=1}^{n_b} (a_{ib} - \hat{\mu}_b)^2$
- Moment conditions:
  - $g_b^{\mu}(\theta) = \hat{\mu}_b - A_b \Gamma\left(1 - \frac{1}{\varepsilon}\right) = 0$
  - $g_b^{\sigma}(\theta) = \hat{\sigma}_b^2 - A_b^2 \left[\Gamma\left(1 - \frac{2}{\varepsilon}\right) - \Gamma\left(1 - \frac{1}{\varepsilon}\right)^2\right] = 0$
- Identification:
  - 20 buildings $\Rightarrow$ 40 empirical moments
    - Mean and variance for each building
  - 21 parameters (ε and 1 A_b per building) $\Rightarrow$ overidentified system
- Estimation method: One-step GMM with identity weighting matrix

Back to Preferences

SMM: Calibration targets

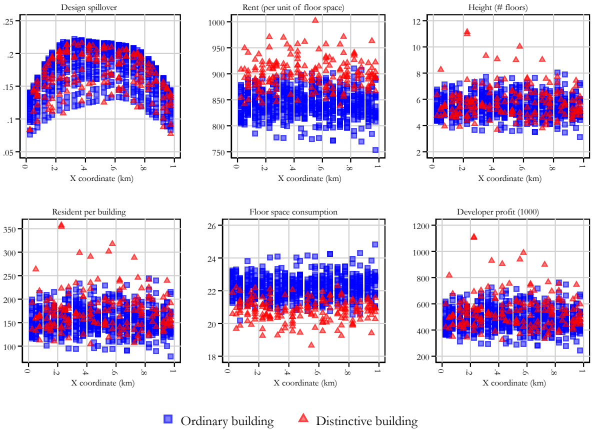
- Parameters to calibrate:
  - Preference for distinctive design: $A^{d=\text{distinctive}}$
  - Average distinctive design cost: $\bar{\delta}^{d=\text{distinctive}}$
- Distinctive design cost: Common component plus shock from normal distribution
  - $\delta_i^{d=\text{distinctive}} = \bar{\delta}^{d=\text{distinctive}} \left(1 + \sigma \frac{\tilde{\delta}_i}{\bar{\delta}^{d=\text{distinctive}}}\right)$
- Empirical target moments:
  - Internal design premium: $\hat{a} = 15\%$
  - Share of distinctive buildings: $\hat{s} = 20\%$
- SMM objective:
  - Match model-implied premium and distinctive share
  - Internal premium identifies preferences
  - Distinctive share identifies costs

SMM: Numerical procedure

- Inner loop:
  - Solve developer design choices and heights
  - Iterate until no parcel switches design
- Middle loop:
  - Update utilities and population shares
  - Clear housing markets across submarkets
- Outer loop:
  - Update structural parameters: $A^{d=\text{distinctive}}$,
    - $\bar{\delta}^{d=\text{distinctive}}$
- Numerical objective:
  - Minimize weighted distance between:
    - Simulated and empirical internal premium
    - Simulated and empirical distinctive share
- Main calibration:
  - $A^{d=\text{distinctive}} = 0.772$, $\bar{\delta}^{d=\text{distinctive}} = 0.219$

Back to Quantification

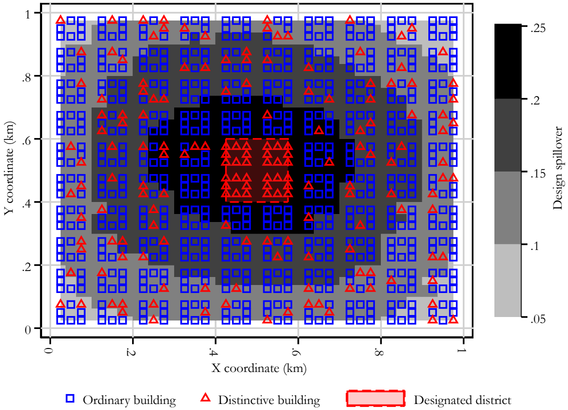
Subsidy Gradients



- Effects across subsidy levels:
  - Distinctive building share rises smoothly
  - Rents increase, but at diminishing rate
- Margins of adjustment:
  - Utility gain from spillovers
  - Developer profits vs. tax burden
- Welfare profile:
  - Concave, consistent with downward-sloping demand
  - Peak defines Pigovian optimum

Back to Pigovian subsidy

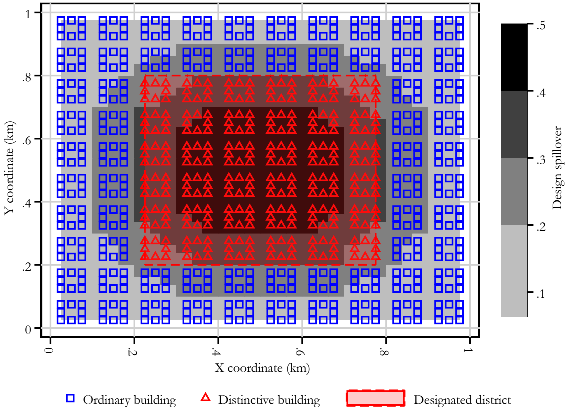
Mandatory District Map



- Visualizes the location of the mandatory design district
- Shows spatial clustering of distinctive buildings
- Useful for interpreting spillover and coordination effects

Back to Mandatory District

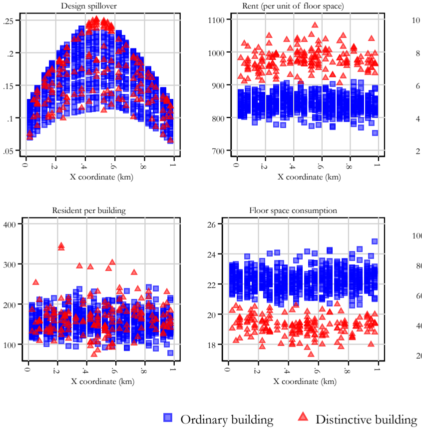
Mandatory District Map



- All distinctive buildings now in the district
- Many more distinctive buildings
- Profitable distinctive buildings **crowded out**

Back to Mandatory District

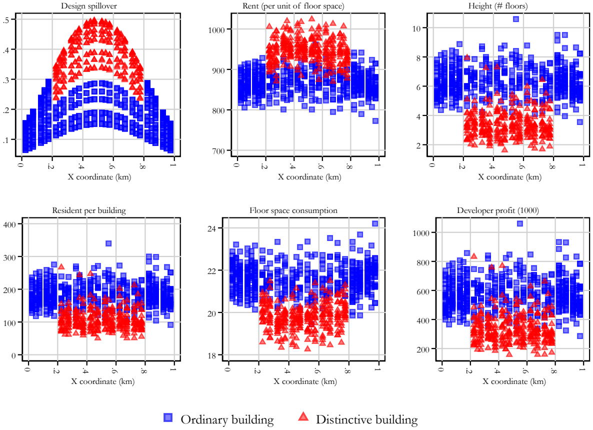
Mandatory District: Gradients



- When the mandatory district is large:
  - Distinctive share saturates
  - Returns diminish, congestion effects rise
- Welfare impact turns negative
  - Utility and profits decline
- Policy trade-off:
  - Small districts solve coordination
  - Large ones impose inefficiency

Back to Mandatory District

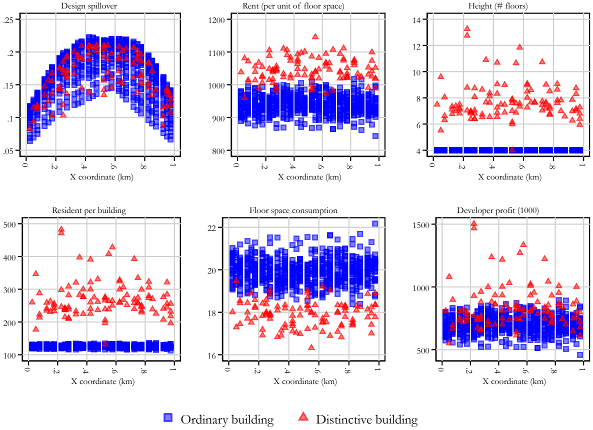
Mandatory District: Large-Scale Scenario



- When the mandatory district is large:
  - Distinctive share saturates
  - Returns diminish, congestion effects rise
- Welfare impact turns negative
  - Utility and profits decline
- Policy trade-off:
  - Small districts solve coordination
  - Large ones impose inefficiency

Back to Mandatory District

FAR incentives: Gradients



- With binding height constraints on ordinary buildings
  - Heights of distinctive buildings increase
  - Positive distinctive design premium persists
  - Some distinctive buildings generate large developer profits

Back to FAR incentives